RESEARCH PROJECT ON DATA INTENSIVE COMPUTING

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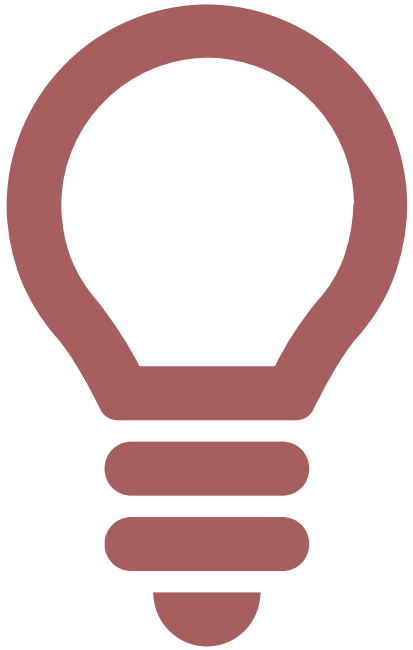
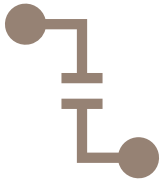
TOPIC 1



ROUTING-AWARE EXPERT REBALANCING FOR MOE TRAINING

MoE models scale large neural networks by activating only a few experts per input



In distributed settings, MoE training has issues:

- expert load imbalance
- heavy cross-node communication

Current MoE frameworks mainly rely on static expert placement, which does not adapt during training

RELEVANCE TO DISTRIBUTED MOE SYSTEMS

Distributed MoE performance depends on:

- where experts are placed and how often experts are used

Static expert placement can lead to:

- overloaded GPUs
- increased all-to-all communication

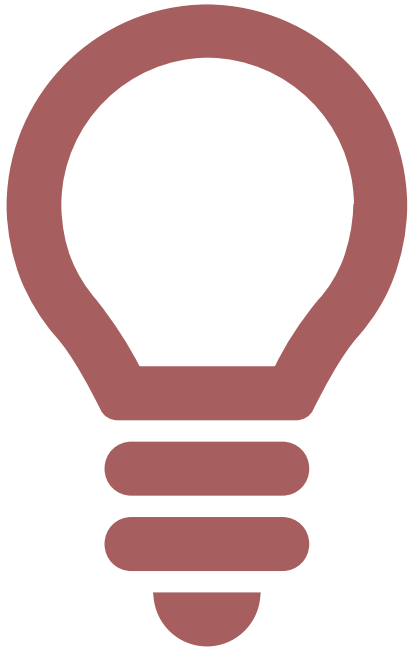
Improving expert placement directly improves:

- Scalability
- Throughput
- resource efficiency

DELIVERABLES

- Working on MoE training on CSC
- Benchmarking and performance analysis
- Routing-aware rebalancing algorithm
- Evaluation against static placement method
- Final report, demo, and presentation

TOPIC 2



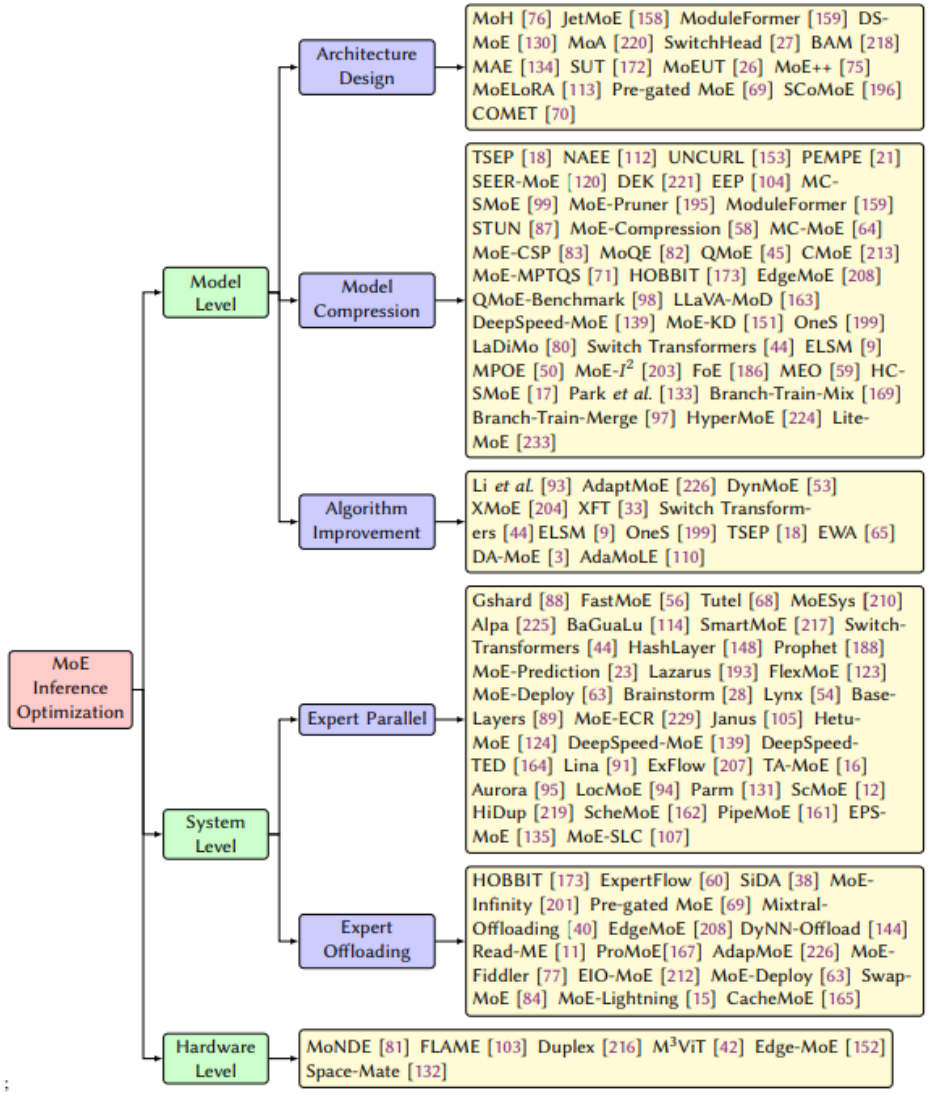


Fig. 1. Taxonomy of MoE inference optimization.

The efficient deployment of MoE models for inference presents unique and significant challenges.

- **Model level**: balancing model capacity with computational efficiency, while routing mechanisms must optimize expert selection and load distribution.
- **System-level**: distributed computation requires sophisticated scheduling algorithms to manage expert placement and activation, load balancing must handle dynamic workload variations across experts, and memory management needs to efficiently handle the loading and unloading of expert parameters.
- **Hardware-level**: mismatch between traditional hardware architectures optimized for dense computation and the sparse, dynamic nature of MoE inference.

Source: Liu, Jiacheng & Tang, Peng & Wang, Wenfeng & Ren, Yuhang & Hou, Xiaofeng & Heng, Pheng-Ann & Guo, Minyi & Li, Chao. (2024). A Survey on Inference Optimization Techniques for Mixture of Experts Models. 10.48550/arXiv.2412.14219.

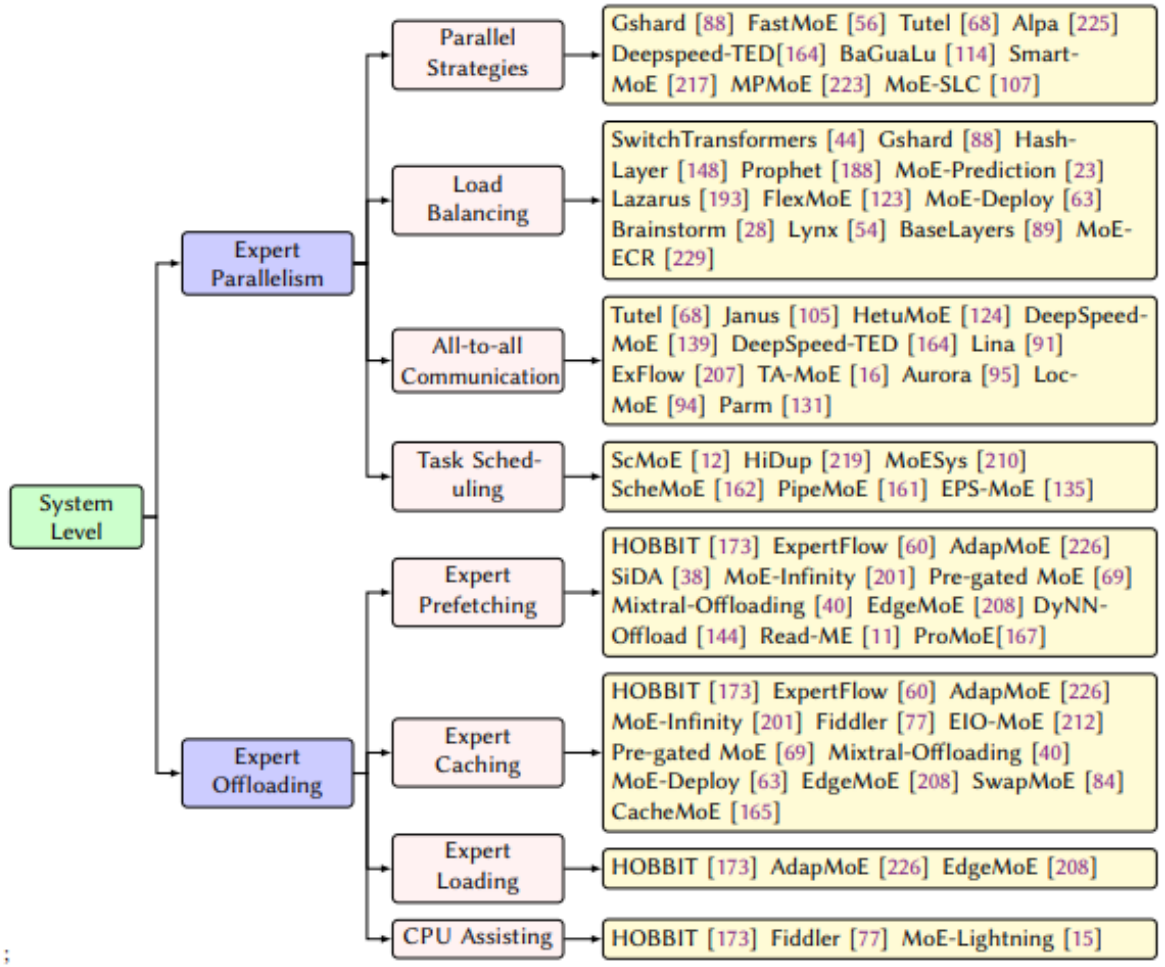
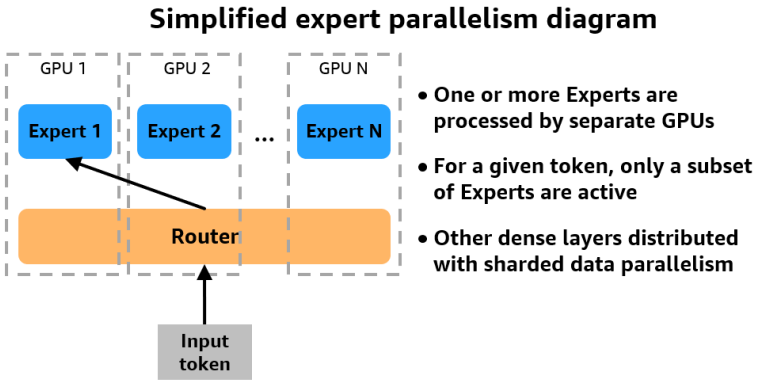


Fig. 6. System-level inference optimization techniques for MoE.

Expert parallelism

key technique for deploying large MoE models across multiple devices to enable distributed execution.



Recent studies combine multiple parallel strategies to maximize parallelism and achieve more efficient distributed computing.

Source: Liu, Jiacheng & Tang, Peng & Wang, Wenfeng & Ren, Yuhang & Hou, Xiaofeng & Heng, Pheng-Ann & Guo, Minyi & Li, Chao. (2024). A Survey on Inference Optimization Techniques for Mixture of Experts Models. 10.48550/arXiv.2412.14219.

HOW IT CAN BE DONE IN THE TEAM

- Combining different parallelism strategies
- Benchmark and performance analysis